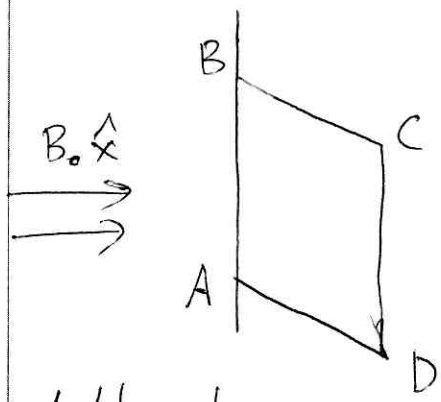
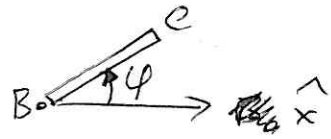
P1



top view:



(a) Let's choose φ to be the angle between x-axis and the frame $\Rightarrow$ then flux through the frame is $\Phi_{ext} = B_0 a^2 \sin\varphi$.

$$L_{mech} = \frac{1}{2} m a^2 \dot{\varphi}^2$$

$$L_c = \frac{1}{2} L \dot{q}^2 = \frac{1}{2} L I^2, \quad I = \dot{q} = \text{current}$$

$$L_{m-e} = I \Phi_{ext} = B_0 a^2 \sin\varphi \, \dot{q}$$

$$L = \frac{1}{2} m a^2 \dot{\varphi}^2 + \frac{1}{2} L \dot{q}^2 + \dot{q} B_0 a^2 \sin\varphi$$

(In part (c) angle φ is introduced as an angle between $\vec{B}$ and the normal to the frame — this is unfortunate but messential mistake on my part. Such an angle is $\tilde{\varphi} = \frac{\pi}{2} + \varphi$ in current notations)

(b) L is t-independent (and no resistance on the circuit) $\Rightarrow$ energy is conserved

$$E = \sum_i \dot{x}_i \frac{\partial L}{\partial \dot{x}_i} - L$$

$$\frac{\partial L}{\partial \dot{q}} = L\dot{q} + \Phi_{ext}, \quad \frac{\partial L}{\partial \dot{\varphi}} = m a^2 \dot{\varphi}$$

$$E = \dot{q}\left(L\dot{q} + \Phi_{ext}\right) + \dot{\varphi}\left(ma^2\dot{\varphi}\right) - \left(\frac{1}{2}ma^2\dot{\varphi}^2 + \frac{1}{2}L\dot{q}^2 + \dot{q}\Phi_{ext}\right)$$

$$= \frac{1}{2}ma^2\dot{\varphi}^2 + \frac{1}{2}L\dot{q}^2 = \text{const.}$$

L also does not depend on charge q $\Rightarrow$

$$p_q = \frac{\partial L}{\partial \dot{q}} = L\dot{q} + \Phi_{ext} = L\dot{q} + B_0 a^2 \sin\varphi = \Phi_0 = \text{total flux}$$

is conserved.

(c) $L\dot{q} + B_0 a^2 \sin\varphi = \Phi_0$

$\dot{q} = \left(\Phi_0 - Ba^2\sin\varphi\right)/L$ (current $\dot{q}$ depends on angle φ!)

$$\Rightarrow E = \frac{1}{2}ma^2\dot{\varphi}^2 + \underbrace{\frac{1}{2L}\left(\Phi_0 - Ba^2\sin\varphi\right)^2}_{U_{eff}(\varphi)}$$

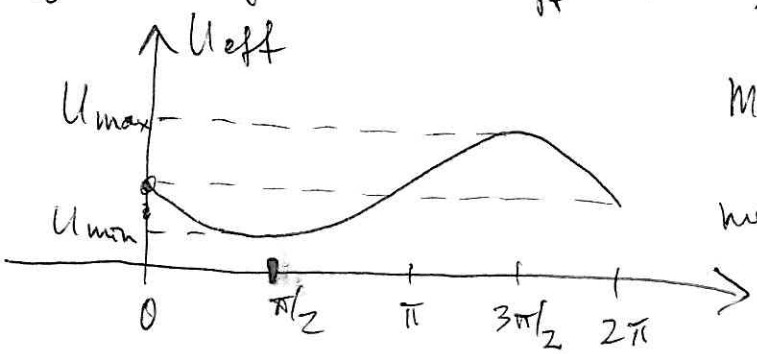
(d) Investigate $U_{eff}(\varphi)$: extrema at $U'_{eff} = 0$

$$\Rightarrow \frac{1}{L}\left(\Phi_0 - Ba^2\sin\varphi\right) Ba^2\cos\varphi = 0$$

$\Rightarrow \cos\varphi = 0$ and $\sin\varphi = \dfrac{\Phi_0}{Ba^2}$ ← only possible if $\Phi_0 < Ba^2$.

$\varphi = \frac{\pi}{2}, \frac{3\pi}{2}$.

1) let $\Phi_0 > Ba^2$ first: U_{eff} has single min at $\varphi = \pi/2$, max at $\varphi = 3\pi/2$.



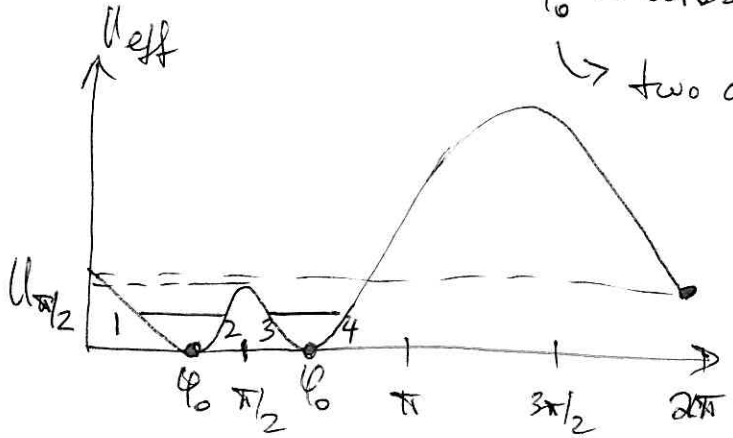
$$\max(U_{eff}) = \frac{1}{2L}\left(\Phi_0 + Ba^2\right)^2$$

$$\min(U_{eff}) = \frac{1}{2L}\left(\Phi_0 - Ba^2\right)^2.$$

2) Let now $\Phi_0 < Ba^2$: then we observe that $\varphi = \pi/2$ becomes local maximum while

$\varphi_0 = \arcsin\left(\frac{\Phi_0}{Ba^2}\right)$ is the minimum $=0$.

↳ two angles actually, to the left and right of $\pi/2$.



denote $U_{\pi/2} = U_{eff}(\pi/2)$
$= \frac{1}{2L}\left(\Phi_0 - Ba^2\right)^2$.

If $E > \max(U_{eff})$ than in both cases (1) and (2) we have rotational motion (with variable angular frequency ~~velocity~~)

If $E < \max(U_{eff})$ than the motion is oscillatory.

If $0 < E < U_{\pi/2} < \max(U_{eff})$ than in case (2) $[\Phi_0 < Ba^2]$ we have oscillatory motion in one of two domains: $\varphi_1 - \varphi_2$ or $\varphi_3 - \varphi_4$.

The turning angles are determined by, for example,

$$E = \frac{1}{2L}\left(\Phi_0 - Ba^2 \sin\varphi_{1,2}\right)^2.$$

Than Period of motion $T_{12} = \sqrt{2ma^2}\int_{\varphi_1}^{\varphi_2} \frac{d\varphi}{\sqrt{E - U_{eff}(\varphi)}}$

for $\varphi_1 < \varphi < \varphi_2$.

The same for $\varphi_3 < \varphi < \varphi_4$.